SepsisSentinel

Calibrated, explainable early warning of ICU sepsis, and what happens when it changes hospitals

Amaan Surti · Data Science & Machine Learning · Semester 7 · October 2026

Why sepsis, why early

- Sepsis = life-threatening organ dysfunction from a dysregulated response to infection (**Sepsis-3**)
- **~49 M cases** and **11 M deaths** worldwide in 2017 (Rudd et al., 2020)
- Each hour of delay in effective antibiotics in septic shock lowers survival (Kumar et al., 2006)
- The problem is **recognition**: early sepsis is subtle, hidden in hourly vitals and sparse labs
- But too many alerts → **alarm fatigue** → alerts ignored

Data: PhysioNet / CinC Challenge 2019, by the numbers

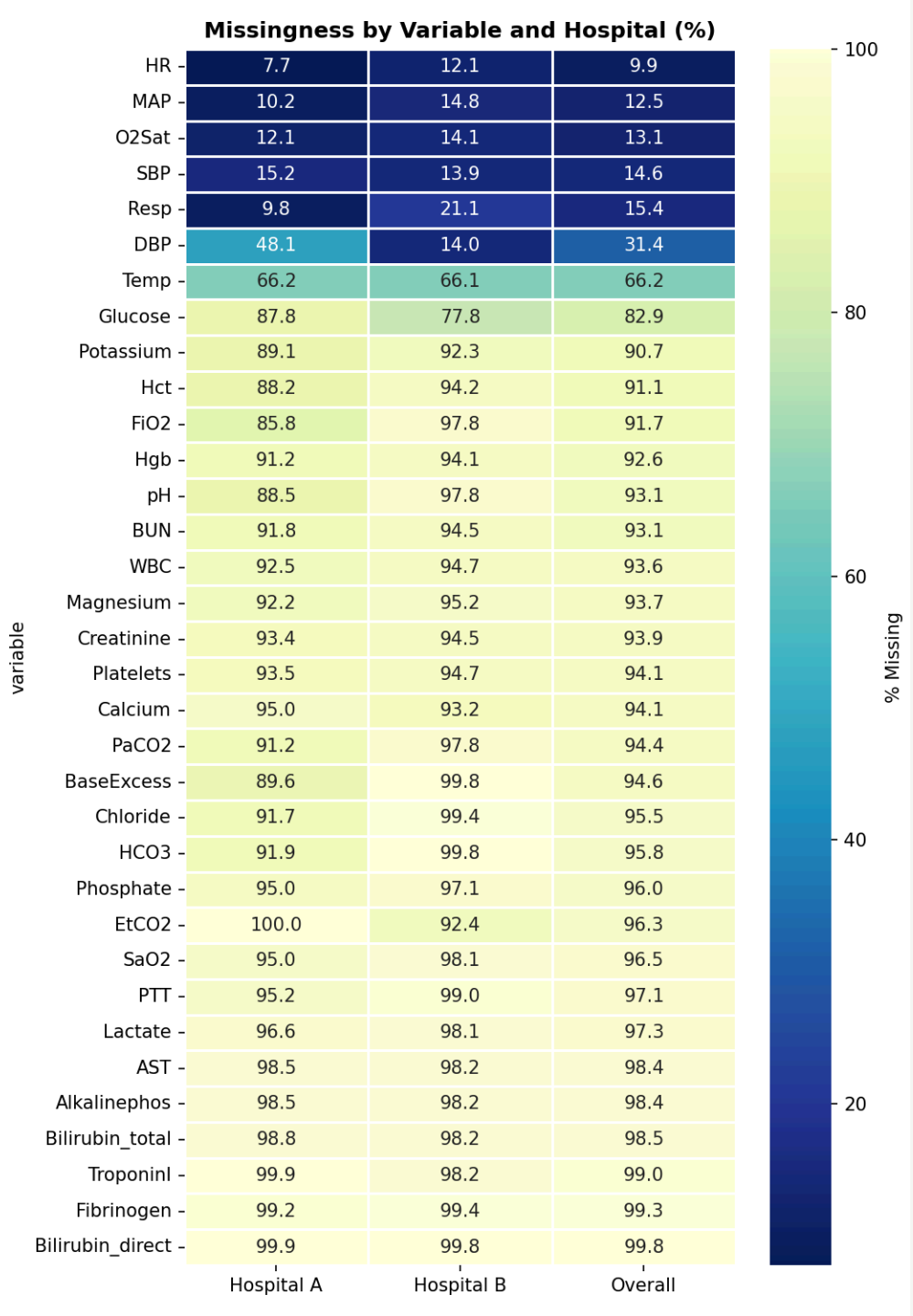
	Hospital A	Hospital B	Total
Patients	20,336	20,000	**40,336**
Hourly rows	790,215	761,995	**1,552,210**
Septic patients	1,790 (8.80%)	1,142 (5.71%)	**2,932 (7.27%)**
Positive hours	2.17%	1.41%	**1.80%**
Median ICU stay	39 h	38 h	38 h

40 variables per hour: 8 vitals, 26 labs, 6 demographic/admin. Label = 1 from **6 h before** Sepsis-3 onset.

Reyna et al., Crit Care Med 2020. Hidden test set and hospital C are not public.

Missing values are the norm, and they differ by hospital

- HR missing in **9.9%** of hours; temperature **66%**; most labs **> 90%**
- DBP missing **48%** at A vs **14%** at B
- An adversarial classifier tells the hospitals apart with **AUROC = 1.0000**, driven mainly by HCO3 **missingness**
- A test that gets **ordered** is a signal of clinical concern



Method: a pipeline that cannot see the future

1. **Patient-level lockbox:** 8,068 test patients locked away; 32,268 dev patients in 5 folds
2. **298 causal features:** LOCF values, measured flags, hours since last test, 6/12/24-h rolling stats & slopes, baseline deltas, shock index, SIRS / partial SOFA
3. **LightGBM**, tuned with Optuna (40 trials), final model 303 rounds
4. **Isotonic calibration** on out-of-fold predictions
5. **Alert policy** chosen to maximise the official utility → **τ = 0.03, k = 1, sticky = 0**
6. **SHAP** reasons for every hour

Leakage test: features on the full stay == features on any prefix (20 random patients) ✓

Evaluation: the Challenge utility rewards timing

- Septic patient: alert in **[t−12, t+3]** earns credit (max +1 from t−6); a missed hour near onset costs up to **−2**
- Non-septic patient: every alert hour costs **−0.05**
- Normalised: never alert = **0**, perfect = **1**
- Also reported: AUROC/AUPRC, calibration (ECE), **sensitivity**, **lead time**, **false alerts per patient-day**, event PPV
- 95% CIs: 1,000 bootstrap resamples of **patients**

Lockbox test (LightGBM): 8,068 patients, evaluated once

0.4270

normalized utility of the LightGBM model (95% CI 0.3929 – 0.4604), the same as its dev out-of-fold estimate (0.4267)

AUROC	Sensitivity	Median lead	False alerts / patient-day	ECE
0.8475	87.54%	29.0 h	0.53	0.0027

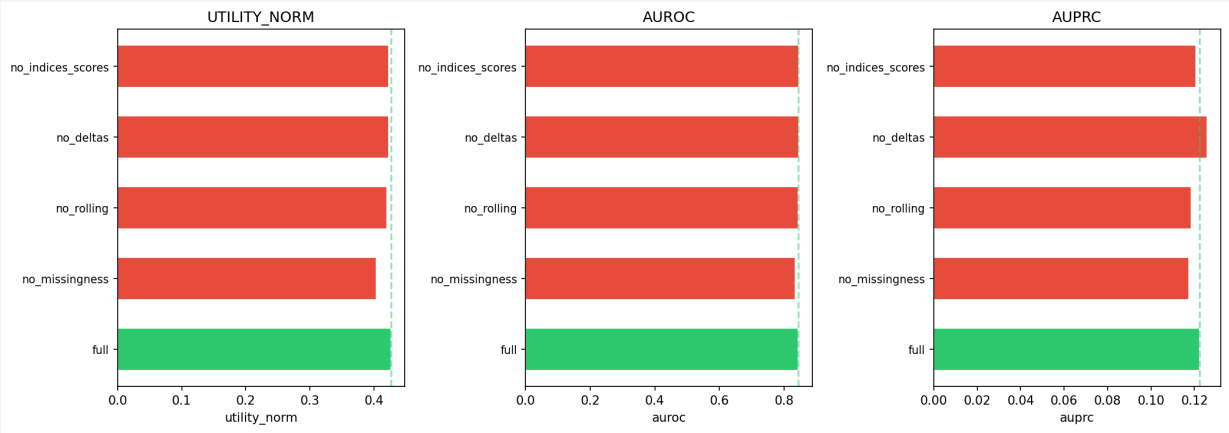
Results vs baselines (same test set, LightGBM)

Model	Utility	AUROC	Sensitivity	False alerts/day
B0 never alert	0.0000	0.500	0.00%	0.00
B1 SIRS ≥ 2	0.0507	0.615	82.76%	**1.53**
B2 qSOFA ≥ 2	0.0122	0.516	**25.94%**	0.45
B3 logistic regression	0.2438	0.737	67.06%	0.25
SepsisSentinel LightGBM	**0.4270**	**0.848**	**87.54%**	0.53

+75% utility over logistic regression · **8.4×** SIRS · AUPRC 0.137 = **7.6×** chance (1.79%)

Ablation: missingness is a signal

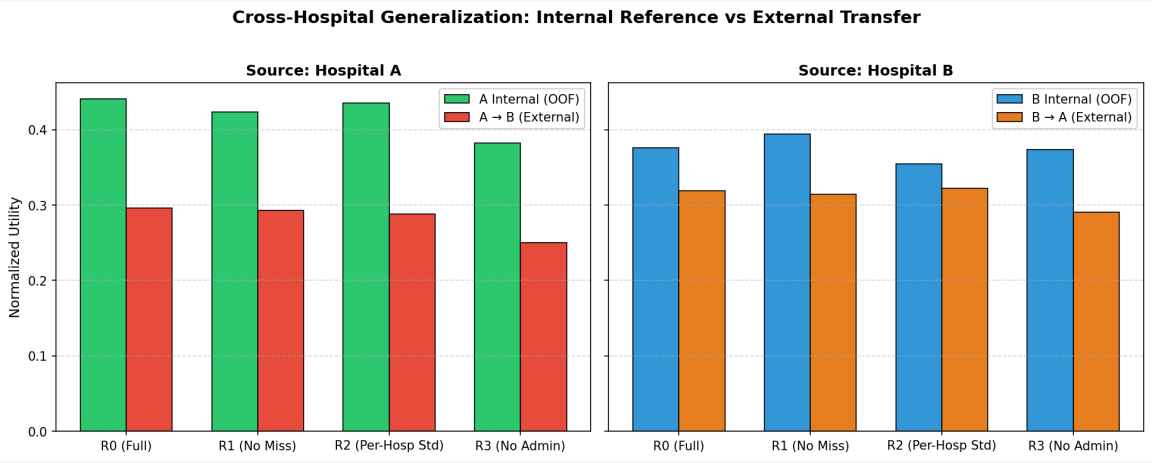
Removed (dev OOF)	Utility	Δ
nothing (298 features)	0.4262	–
missingness F2–F4	**0.4039**	**−0.0223**
rolling stats F5	0.4205	−0.0057
diffs + deltas F6–F7	0.4231	−0.0030
indices + scores F8–F9	0.4234	−0.0028



Cross-hospital finding: performance is partly local

(R0, full features)	Utility	Sens.	FA/day
A internal	0.4405	84.7%	0.54
A → B	**0.2963**	**66.3%**	0.29
B internal	0.3759	76.6%	0.33
B → A	**0.3188**	91.6%	**0.93**

- A → B utility **−32.7%**; AUROC drops only 0.03–0.08
- Main failure = **operating point**: too few alerts (A→B), 3× false alarms (B→A)
- No missingness (R1) or per-site z-scores (R2): gap **not** closed



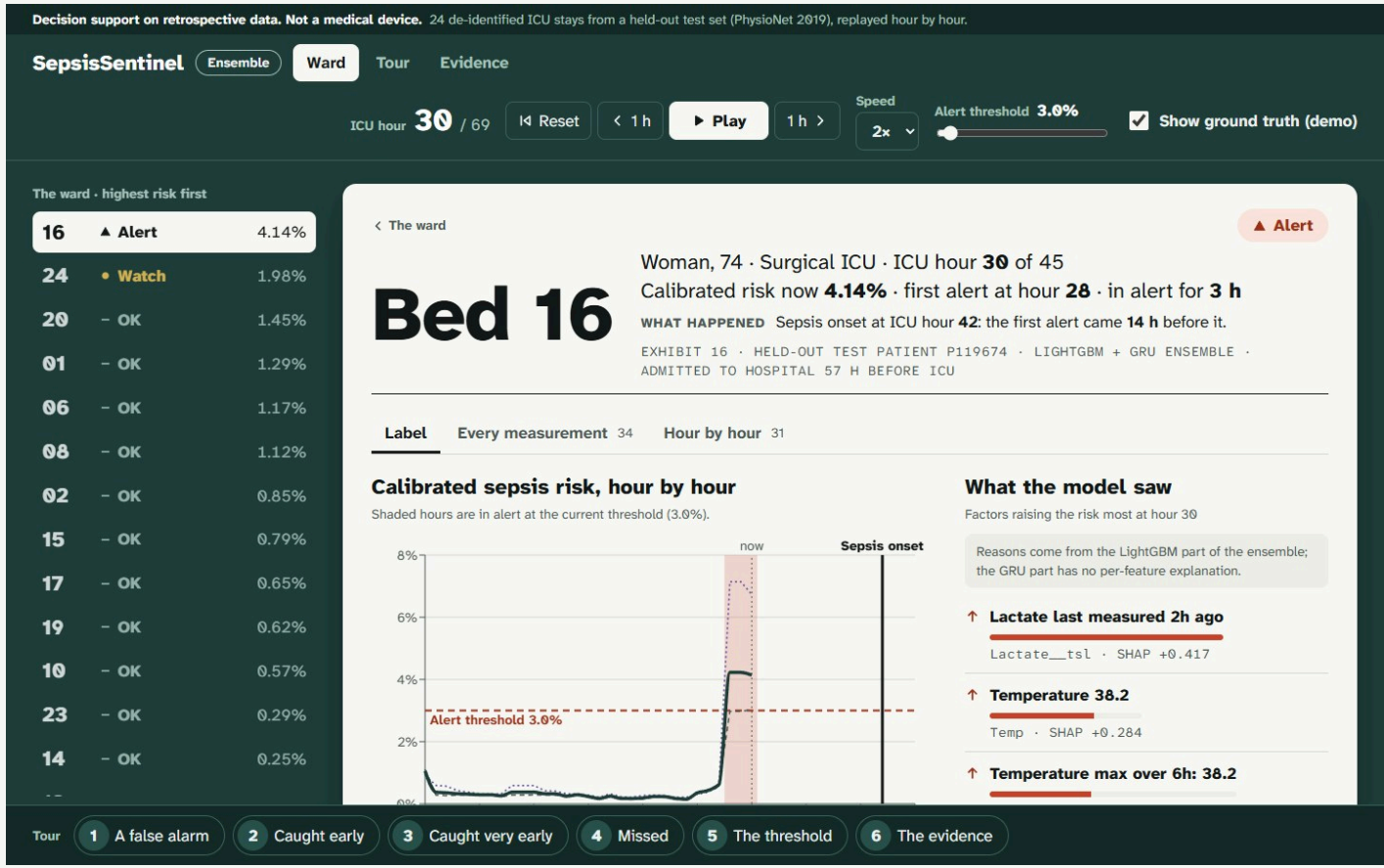
Development comparison (nested 5-fold CV) — not lockbox

32,268 dev patients, nested	Utility	AUROC	Sens.	FA/day
LightGBM	0.4196	0.842	85.5%	0.48
GRU, unidirectional (hidden 64)	0.3529	0.812	78.1%	0.40
Ensemble 0.7 × LightGBM + 0.3 × GRU	**0.4327**	**0.847**	**85.0%**	**0.41**

- Paired patient bootstrap: ensemble − LightGBM **+0.0131 [+0.0072, +0.0190]**
- Search fixed in advance: w ∈ {0.5 … 0.9} × 3 GRU variants; each fold's weight and policy fit on the other 4 folds; all 15 configurations reported
- **Served model = this ensemble. Its lockbox score has not been measured; 0.4270 belongs to LightGBM alone.**

Product: from model to bedside demo

- **FastAPI** serves the **ensemble** (default) or LightGBM (`MODEL_KIND`); GRU on CPU
- **React + TypeScript** dashboard in a museum-gallery world: a wall of numbered beds, each opening onto an object label with all 34 measurements and an insight for every hour; tour rail of six stops
- One code path for training and online `/api/predict` (diff < 10^{-6}; ensemble online vs batch 4×10^{-7})
- **Docker Compose**: API :8000 + UI :8080
- 93 automated tests, incl. causality of the full ensemble path



Live demo (3 minutes)

1. Ward at hour 0 (badge: **Ensemble**) → **Play** (Space, 2×)
2. Pause when **Bed 03** turns **Alert** (hour 16), the first septic bed to alert
3. Bed 03 mounts on its label → read **"What the model saw"**
4. Toggle **Show ground truth** → onset at hour 22, 6 h after the alert
5. Show a miss (**Bed 07**), move the **threshold slider** → more / fewer alerts; Evidence: lockbox room vs development annex
6. Close on the disclaimer

Fallback: 60-s screen recording in case the laptop fails.

Limitations and ethics

- Retrospective, de-identified data from two hospitals; **no hospital C**, external validity only simulated
- Sepsis-3 labels from records are not perfect ground truth
- $\tau = 0.03 \rightarrow$ 20% of hours in alert, event PPV 4.6%: sensitivity bought with alerts
- Lead time inflated by persistent early alerts; utility gives no credit before t−12
- SHAP explains the **model**, not causes; timing features reflect clinician behaviour
- Subgroups underpowered (weakest: age ≥ 80, sensitivity 80.3%)
- Served ensemble: **no lockbox estimate**; its gain is development-only, reasons cover the LightGBM half
- **Not a medical device**

Takeaways

1. A causal, calibrated LightGBM reaches **utility 0.427**, **AUROC 0.848**, **87.5%** sensitivity on a once-only test set
2. It beats SIRS, qSOFA and logistic regression by a wide margin at a known alarm cost (**0.53/day**)
3. **Missingness carries signal**, and makes the model partly hospital-specific
4. Moving hospitals costs up to **33%** utility, mostly through the alert threshold
5. A causal GRU helps only as a partner: ensemble **+0.013** dev utility (nested CV), served but not lockbox-tested
6. Validation at your own hospital is where it starts, not where it ends

Thank you. Questions?